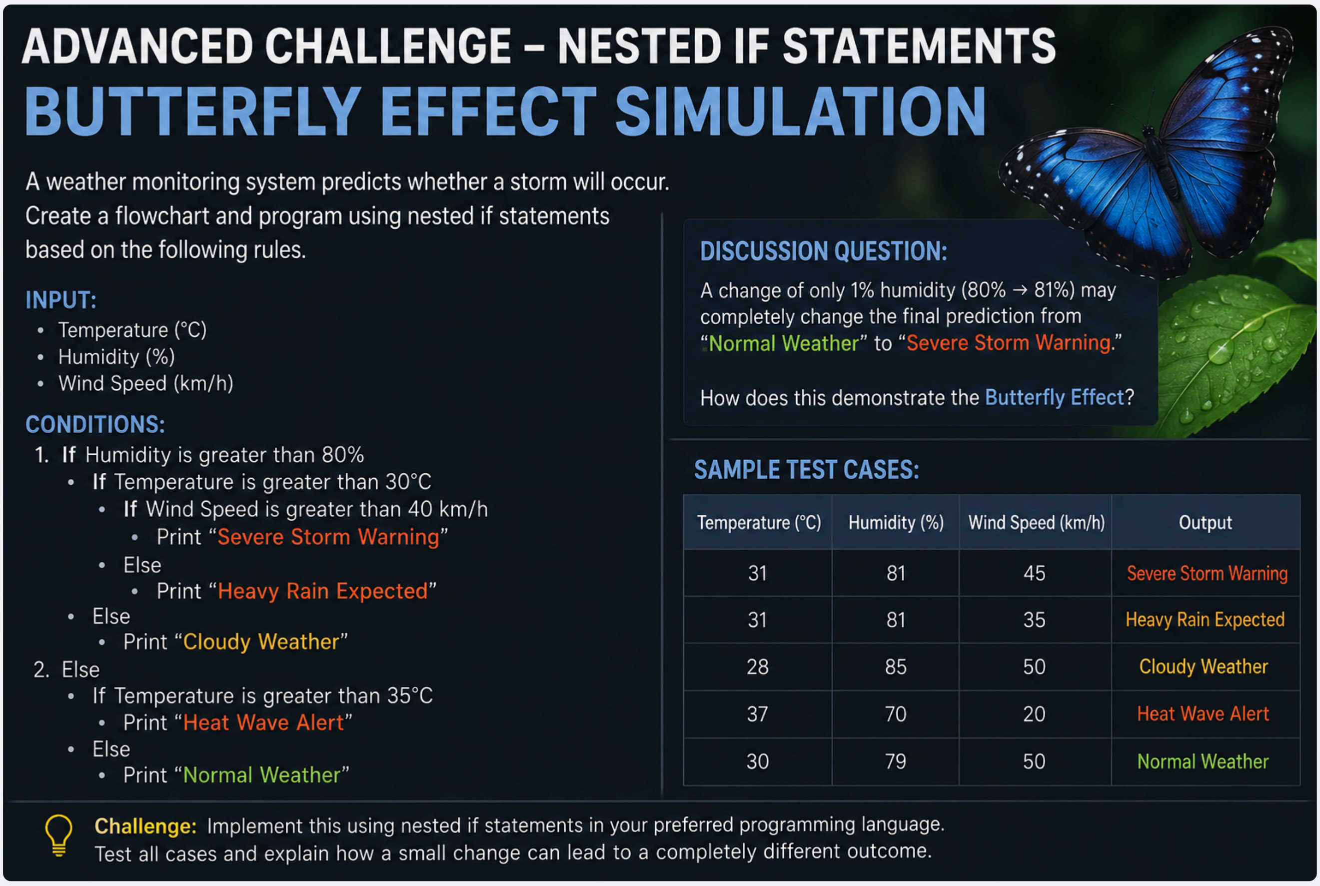
challenge 260604

Challenge



ADVANCED CHALLENGE – NESTED IF STATEMENTS

BUTTERFLY EFFECT SIMULATION

A weather monitoring system predicts whether a storm will occur. Create a flowchart and program using nested if statements based on the following rules.

INPUT:
- Temperature (°C)
- Humidity (%)
- Wind Speed (km/h)

CONDITIONS:
1. **If** Humidity is greater than 80%
    - **If** Temperature is greater than 30°C
        - **If** Wind Speed is greater than 40 km/h
            - Print "Severe Storm Warning"
        - Else
            - Print "Heavy Rain Expected"
    - Else
        - Print "Cloudy Weather"
2. Else
    - If Temperature is greater than 35°C
        - Print "Heat Wave Alert"
    - Else
        - Print "Normal Weather"

DISCUSSION QUESTION:

A change of only 1% humidity (80% → 81%) may completely change the final prediction from "Normal Weather" to "Severe Storm Warning."

How does this demonstrate the Butterfly Effect?

SAMPLE TEST CASES:

Temperature (°C)	Humidity (%)	Wind Speed (km/h)	Output
31	81	45	Severe Storm Warning
31	81	35	Heavy Rain Expected
28	85	50	Cloudy Weather
37	70	20	Heat Wave Alert
30	79	50	Normal Weather

Challenge: Implement this using nested if statements in your preferred programming language. Test all cases and explain how a small change can lead to a completely different outcome.

Python code

```python
temperature = float(input("Enter the temperature (°C): "))
humidity = float(input("Enter the humidity (%): "))
wind_speed = float(input("Enter wind speed (km/h): "))

if humidity > 80:
      if temperature > 30:
            if wind_speed > 40:
                  print("Severe Storm Warning")
            else:
                  print("Heavy rain expected")
      else:
            print("Cloudy weather")
else:
      if temperature > 35:
            print("Heat wave alert")
      else:
            print("Normal weather")
```

Butterfly Effect & Chaos Theory

- **Butterfly Effect**
  - the idea that small, seemingly insignificant actions can cause massive, unpredictable consequences downstream
  - foundational principle of Chaos Theory
- **Chaos Theory**
  - the branch of mathematics and physics that deals with systems that look completely random, unpredictable, and messy on the surface, but are actually governed by strict underlying laws
- If the Butterfly Effect is the concept (small changes cause big results), Chaos Theory is the entire field of study dedicated to figuring out how and why that happens